

第二节 函数的求导法则

一、函数的和、差、积、商的求导法则

二、反函数的求导法则

三、复合函数的求导法则

四、基本求导法则与导数公式

思路:

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} \quad (\text{构造性定义})$$



$$\left\{ \begin{array}{l} (c)' = 0 \\ (\sin x)' = \cos x \\ (\ln x)' = \frac{1}{x} \end{array} \right\} \quad \left. \begin{array}{l} \text{证明中利用了} \\ \text{两个重要极限} \end{array} \right\}$$

本节内容



求导法则

其他基本初等
函数求导公式

初等函数求导问题

一、函数的和、差、积、商的求导法则

定理1 如果函数 $u(x), v(x)$ 在点 x 处可导, 则它们的和、差、积、商(分母不为零)在点 x 处也可导, 并且

$$(1) [u(x) \pm v(x)]' = u'(x) \pm v'(x);$$

$$(2) [u(x)v(x)]' = u'(x)v(x) + u(x)v'(x);$$

$$(3) \left[\frac{u(x)}{v(x)} \right]' = \frac{u'(x)v(x) - u(x)v'(x)}{v^2(x)} \quad (v(x) \neq 0).$$

下面只给出(3)的证明, (1)和(2)的证明略.

证 (3) $\left[\frac{u(x)}{v(x)}\right]' = \frac{u'(x)v(x) - u(x)v'(x)}{v^2(x)} \quad (v(x) \neq 0).$

设 $f(x) = \frac{u(x)}{v(x)}, (v(x) \neq 0),$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{u(x+h)}{v(x+h)} - \frac{u(x)}{v(x)}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{u(x+h)v(x) - u(x)v(x+h)}{v(x+h)v(x)h}$$

$$= \lim_{h \rightarrow 0} \frac{u(x+h)v(x) - \cancel{u(x)v(x)} - u(x)v(x+h) + \cancel{u(x)v(x)}}{v(x+h)v(x)h}$$

$$= \lim_{h \rightarrow 0} \frac{[u(x+h) - u(x)]v(x) - u(x)[v(x+h) - v(x)]}{v(x+h)v(x)h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{u(x+h) - u(x)}{h} \cdot v(x) - u(x) \cdot \frac{v(x+h) - v(x)}{h}}{v(x+h)v(x)h}$$

$$= \frac{u'(x)v(x) - u(x)v'(x)}{[v(x)]^2},$$

$\therefore f(x)$ 在 x 处可导. 证毕

推论

$$(1) [f_1(x) + f_2(x) + \cdots + f_s(x)]' = f_1'(x) + f_2'(x) + \cdots + f_s'(x);$$

$$(2) [Cf(x)]' = Cf'(x);$$

$$(3) [f_1(x) f_2(x) \cdots f_s(x)]' = f_1'(x) f_2(x) \cdots f_s(x) + \\ f_1(x) f_2'(x) \cdots f_s(x) + \\ \cdots + \\ f_1(x) f_2(x) \cdots f_s'(x).$$

例1 求 $y = 2x^3 - 5x^2 + 3x - 7$ 的导数.

解 $y' = 6x^2 - 10x + 3.$

例2 $f(x) = x^3 + 4\cos x - \sin \frac{\pi}{2}$, 求 $f'(x)$ 及 $f'(\frac{\pi}{2})$.

解 $f'(x) = 3x^2 - 4 \sin x,$

$$f'(\frac{\pi}{2}) = \frac{3}{4}\pi^2 - 4.$$

例3 $y=e^x(\sin x + \cos x)$, 求 y' .

解

$$\begin{aligned} y' &= (e^x)'(\sin x + \cos x) + e^x(\sin x + \cos x)' \\ &= e^x(\sin x + \cos x) + e^x(\cos x - \sin x) = 2e^x \cos x. \end{aligned}$$

例4 求 $y=\tan x$ 的导数.

解

$$\begin{aligned} y' &= (\tan x)' = \left(\frac{\sin x}{\cos x} \right)' = \frac{(\sin x)' \cos x - \sin x (\cos x)'}{\cos^2 x} \\ &= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x, \end{aligned}$$

即 $(\tan x)' = \sec^2 x$.

同理可得 $(\cot x)' = -\csc^2 x$.

例5 求 $y = \sec x$ 的导数.

解
$$y' = (\sec x)' = \left(\frac{1}{\cos x} \right)' = \frac{1' \cdot (\cos x) - 1 \cdot (\cos x)'}{\cos^2 x} = \frac{\sin x}{\cos^2 x} = \sec x \tan x,$$

即 $(\sec x)' = \sec x \tan x.$

同理可得 $(\csc x)' = -\csc x \cot x.$

二、反函数的导数

定理2 如果函数 $x = f(y)$ 在某区间 I_y 内单调、可导且 $f'(y) \neq 0$, 则它的反函数 $y = f^{-1}(x)$ 在区间 $I_x = \{x | x = f(y), y \in I\}$ 内也可导, 且有

$$[f^{-1}(x)]' = \frac{1}{f'(y)} \text{ 或 } \frac{dy}{dx} = \frac{1}{\frac{dx}{dy}}.$$

证 由定理条件可知,

反函数 $y = f^{-1}(x)$ 存在且在 I_x 内单调、连续.

任取 $x \in I_x$, 给 x 以增量 $\Delta x \neq 0$, 则 $\Delta y \neq 0$, 且 $\lim_{\Delta x \rightarrow 0} \Delta y = 0$.

$$\therefore [f^{-1}(x)]' = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta y \rightarrow 0} \frac{1}{\frac{\Delta x}{\Delta y}} = \frac{1}{\lim_{\Delta y \rightarrow 0} \frac{\Delta x}{\Delta y}} = \frac{1}{f'(y)}.$$

证毕

即 反函数的导数等于直接函数导数的倒数.

例6 求函数 $y = \arcsin x$ 的导数.

解 $\because x = \sin y$ 在 $I_y = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ 内单调、可导, 且 $(\sin y)' = \cos y > 0$,

$\therefore$ 在 $I_x = (-1, 1)$ 内有

$$(\arcsin x)' = \frac{1}{(\sin y)'} = \frac{1}{\cos y} = \frac{1}{\sqrt{1 - \sin^2 y}} = \frac{1}{\sqrt{1 - x^2}}.$$

$$\text{即 } (\arcsin x)' = \frac{1}{\sqrt{1 - x^2}}.$$

同理可得 $(\arccos x)' = -\frac{1}{\sqrt{1 - x^2}}.$

例7 求函数 $y = \arctan x$ 的导数.

解

$\because x = \tan y$ 在 $I_y = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ 内单调、可导,

且 $(\tan y)' = \sec^2 y \neq 0$, $\therefore$ 在 $I_x = (-\infty, +\infty)$ 内有

$$(\arctan x)' = \frac{1}{(\tan y)'} = \frac{1}{\sec^2 y} = \frac{1}{1 + \tan^2 y} = \frac{1}{1 + x^2}.$$

$$\text{即 } (\arctan x)' = \frac{1}{1 + x^2};$$

同理可得 $(\operatorname{arccot} x)' = -\frac{1}{1 + x^2}.$

例8 求函数 $y = \log_a x$ 的导数.

解

$\because x = a^y$ 在 $I_y = (-\infty, +\infty)$ 内单调、可导,

且 $(a^y)' = a^y \ln a \neq 0$, $\therefore$ 在 $I_x = (0, +\infty)$ 内有,

$$(\log_a x)' = \frac{1}{(a^y)'} = \frac{1}{a^y \ln a} = \frac{1}{x \ln a}.$$

$$\text{即 } (\log_a x)' = \frac{1}{x \ln a};$$

$$\text{特别地 } (\ln x)' = \frac{1}{x}.$$

补例： $f(x) = 2x + \cos x$, $\varphi(1) = f^{-1}(1)$, 求 $\varphi'(1)$.

解： 令 $f(x) = y$, 则 $\varphi'(y) = \frac{1}{f'(x)}$, 而 $f(0) = 1$, 故而

$$\varphi'(1) = \frac{1}{f'(0)} = \frac{1}{(2x + \cos x)'|_{x=0}} = \frac{1}{2}.$$

课后练习： 设 $f(x) = \ln x + \arctan x$, $g(x) = f^{-1}(x)$, 求 $g'\left(\frac{\pi}{4}\right)$. $\frac{2}{3}$



三、复合函数的求导法则

定理3 如果函数 $u=g(x)$ 在点 x 可导, 而 $y=f(u)$ 在点 $u=g(x)$ 可导, 那么复合函数 $y=f[g(x)]$ 在点 x 可导, 且其导数为

$$\frac{dy}{dx} = f'(u) \cdot g'(x).$$

证 $\because y=f(u)$ 在点 u 可导, $\therefore \lim_{\Delta u \rightarrow 0} \frac{\Delta y}{\Delta u} = f'(u).$

$\therefore \frac{\Delta y}{\Delta u} = f'(u) + \alpha \quad (\lim_{\Delta u \rightarrow 0} \alpha = 0),$ 于是 $\Delta y = f'(u)\Delta u + \alpha\Delta u,$

故 $\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \left[f'(u) \frac{\Delta u}{\Delta x} + \alpha \frac{\Delta u}{\Delta x} \right] = f'(u)g'(x).$ **证毕**

链式法则

$$y = f[g(x)]: y = f(u), u = g(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = f'(u) \cdot g'(x)$$

$$y \text{ — } u \text{ — } x$$

y

|

u

|

v

|

x

推广

$$y = f(g(h(x))): y = f(u), u = g(v), v = h(x).$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx} = f'(u) \cdot g'(v) \cdot h'(x)$$

因变量对自变量求导, 等于因变量对中间变量求导, 乘以中间变量对自变量求导.

关键: 搞清复合函数结构, 由外向内逐层求导.

例9 设 $y = e^{x^3}$, 求 $\frac{dy}{dx}$.

解 $\because y = e^u, u = x^3. \therefore \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = e^u \cdot 3x^2 = 3x^2 e^{x^3}.$

例10 设 $y = (x^2+1)^{100}$, 求 $\frac{dy}{dx}$.

解 $\because y = u^{100}, u = x^2+1.$

$$\therefore \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = 100 \cdot u^{99} \cdot (2x) = 200 x (x^2+1)^{99}.$$

例11 设 $y = \sin \frac{2x}{1+x^2}$, 求 $\frac{dy}{dx}$.

解 $\because y = \sin u, u = \frac{2x}{1+x^2}$.

$$\text{而 } \frac{dy}{du} = \cos u, \frac{du}{dx} = \frac{2 \cdot (1+x^2) - 2x \cdot 2x}{(1+x^2)^2} = \frac{2 \cdot (1-x^2)}{(1+x^2)^2}.$$

$$\therefore \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \cos u \cdot \frac{2(1-x^2)}{(1+x^2)^2} = \frac{2(1-x^2)}{(1+x^2)^2} \cos \frac{2x}{1+x^2}.$$



课堂练习

设 $y = \sqrt[3]{1 - 2x^2}$, 求 $\frac{dy}{dx}$.

答案 $\because y = \sqrt[3]{u} = u^{\frac{1}{3}}, u = 1 - 2x^2$.

$$\begin{aligned}\therefore \frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} = \frac{1}{3} \cdot u^{-\frac{2}{3}} \cdot (-4x) \\ &= \frac{1}{3} \cdot (1 - 2x^2)^{-\frac{2}{3}} \cdot (-4x) \\ &= \frac{-4x}{3\sqrt[3]{(1 - 2x^2)^2}}.\end{aligned}$$

熟悉后, 不写出中间变量

解 $\because y = \sqrt[3]{1 - 2x^2} = (1 - 2x^2)^{\frac{1}{3}},$

$$\begin{aligned}\therefore y' &= \frac{1}{3} \cdot (1 - 2x^2)^{-\frac{2}{3}} \cdot (1 - 2x^2)' \\ &= \frac{1}{3} \cdot (1 - 2x^2)^{-\frac{2}{3}} \cdot (-4x) \\ &= \frac{-4x}{3\sqrt[3]{(1 - 2x^2)^2}}.\end{aligned}$$

例12 设 $y = \ln \cos(\sin x)$, 求 $\frac{dy}{dx}$.

解

$$\because y = \ln u, u = \cos v, v = \sin x,$$

$$\begin{aligned}\therefore \frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx} \\ &= \frac{1}{u} \cdot (-\sin v) \cdot \cos x \\ &= \frac{\sin(\sin x)}{\cos(\sin x)} \cos x. \\ &= -\tan(\sin x) \cdot \cos x.\end{aligned}$$

熟悉后, 不写出中间变量

解

$$\because y = \ln \cos(\sin x)$$

$$\begin{aligned}\therefore y' &= \frac{1}{\cos(\sin x)} \cdot (\cos(\sin x))' \\ &= \frac{1}{\cos(\sin x)} (-\sin(\sin x)) \cdot (\sin x)' \\ &= -\tan(\sin x) \cdot \cos x.\end{aligned}$$

例13 设 $y = e^{\sin \frac{1}{x}}$, 求 $\frac{dy}{dx}$.

解

$$\because y = e^u, u = \sin v, v = \frac{1}{x},$$

$$\begin{aligned}\therefore \frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx} \\ &= e^u \cdot \cos v \cdot \left(-\frac{1}{x^2}\right) \\ &= e^{\sin \frac{1}{x}} \cdot \cos \frac{1}{x} \cdot \left(-\frac{1}{x^2}\right).\end{aligned}$$

熟悉后, 不写出中间变量

解 $\because y = e^{\sin \frac{1}{x}},$

$$\begin{aligned}\therefore y' &= e^{\sin \frac{1}{x}} \cdot \left(\sin \frac{1}{x}\right)' \\ &= e^{\sin \frac{1}{x}} \cdot \cos \frac{1}{x} \cdot \left(\frac{1}{x}\right)' \\ &= e^{\sin \frac{1}{x}} \cdot \cos \frac{1}{x} \cdot \left(-\frac{1}{x^2}\right).\end{aligned}$$

练习：设 $f(x) = \ln \sin(1 + \sqrt{x})$, 求 $f'(x)$.

$$\frac{1}{2\sqrt{x}} \cot(1 + \sqrt{x})$$

四、初等函数的求导问题——初等函数的导数仍为初等函数

1. 常数和基本初等函数的导数公式 —熟练记忆

$$(1) (c)' = 0$$

$$(2) (x^\mu)' = \mu x^{\mu-1}$$

$$(3) (\sin x)' = \cos x$$

$$(4) (\cos x)' = -\sin x$$

$$(5) (\tan x)' = \sec^2 x$$

$$(6) (\cot x)' = -\csc^2 x$$

$$(7) (\sec x)' = \sec x \tan x$$

$$(8) (\csc x)' = -\csc x \cot x$$

$$(9) (a^x)' = a^x \ln a$$

$$(a > 0, a \neq 1)$$

$$(10) (e^x)' = e^x$$

$$(11) (\log_a x)' = \frac{1}{x \ln a}$$

$$(a > 0, a \neq 1)$$

$$(12) (\ln x)' = \frac{1}{x}$$

$$(13) (\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$$

$$(14) (\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$$

$$(15) (\arctan x)' = \frac{1}{1+x^2}$$

$$(16) (\operatorname{arccot} x)' = -\frac{1}{1+x^2}$$

2. 函数的和、差、积、商的求导法则

设 $u=u(x)$, $v=v(x)$ 都可导, 则

$$(1) (u \pm v)' = u' \pm v'; \quad (2) (Cu(x))' = Cu'(x) (C \text{ 是常数});$$

$$(3) (uv)' = u'v + uv'; \quad (4) \left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2} \quad (v \neq 0).$$

3. 复合函数的求导法则

设 $y=f(u)$, 而 $u=g(x)$ 且 $f(u)$ 及 $g(x)$ 都可导,

则复合函数 $y=f[g(x)]$ 的导数为

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \text{ 或 } y'(x) = f'(u) \cdot g'(x).$$

4. 反函数的求导法则

如果函数 $x = f(y)$ 在某区间 I_y 内单调、可导且 $f'(y) \neq 0$, 则它的反函数 $y = f^{-1}(x)$ 在区间 $I_x = \{x | x = f(y), y \in I\}$ 内也可导, 且有

$$[f^{-1}(x)]' = \frac{1}{f'(y)} \text{ 或 } \frac{dy}{dx} = \frac{1}{\frac{dx}{dy}}.$$

利用上述公式及法则初等函数求导问题可完全解决.

5. 初等函数求导举例

例14 设 $y = \sin nx \cdot \sin^n x$ (n 为常数), 求 y' .

解

$$\begin{aligned} y' &= (\sin nx \cdot \sin^n x)' \\ &= (\sin nx)' \cdot \sin^n x + (\sin nx) \cdot (\sin^n x)' \\ &= (n \cdot \cos nx) \cdot \sin^n x + (\sin nx) \cdot (n \sin^{n-1} x \cdot \cos x) \\ &= n \sin^{n-1} x (\cos nx \cdot \sin x + \sin nx \cdot \cos x) \\ &= n \sin^{n-1} x \sin(n+1)x. \end{aligned}$$

例15 设函数 $f(x) = \ln(x + \sqrt{1 + x^2})$, 求 y' .

解

$$y' = \frac{1}{x + \sqrt{1 + x^2}} \cdot (x + \sqrt{1 + x^2})'$$

$$= \frac{1}{x + \sqrt{1 + x^2}} \cdot \left(1 + \frac{2x}{2\sqrt{1 + x^2}}\right)$$

$$= \frac{1}{\sqrt{1 + x^2}}.$$

$$(\sqrt{1 + x^2})' = \frac{2x}{2\sqrt{1 + x^2}}$$

例16 设函数 $f(x)$ 和 $g(x)$ 可导, 且 $f^2(x)+g^2(x)\neq 0$, 试求函数

$y = \sqrt{f^2(x)+g^2(x)}$ 的导数.

解

$$y' = \frac{1}{2\sqrt{f^2(x)+g^2(x)}} \cdot (f^2(x)+g^2(x))' \quad (\sqrt{x})' = \frac{1}{2\sqrt{x}}$$

$$= \frac{1}{2\sqrt{f^2(x)+g^2(x)}} \cdot (2f(x) \cdot f'(x) + 2g(x) \cdot g'(x))$$

$$= \frac{f(x) \cdot f'(x) + g(x) \cdot g'(x)}{\sqrt{f^2(x)+g^2(x)}}.$$

思考：设 $f(x)$ 可导，求 $y = f(\sin^2 x) + f(e^{\sqrt{x}})$ 的导数.

思考：设 $y = f\left(\frac{3x-2}{3x+2}\right)$ ，且 $f'(x) = \arccos x^2$ ，求 $\frac{dy}{dx} \big|_{x=0}$. 0

补例：设 $f(x) = \begin{cases} e^{ax}, & x \leq 0 \\ b + \sin 2x, & x > 0 \end{cases}$ ， a, b 为何值时， $f(x)$ 可导，

并写出其导数.

$$a = 2, b = 1, f'(x) = \begin{cases} 2e^{2x}, & x < 0 \\ 2, & x = 0 \\ 2\cos x, & x > 0 \end{cases}$$

补例：讨论函数 $f(x) = \begin{cases} x^\alpha \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ 在 $x = 0$ 处可导性以及导函数的连续性.

$$\alpha > 1, f'(x) = \begin{cases} \alpha x^{\alpha-1} \sin \frac{1}{x} - x^{\alpha-2} \cos \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

注：当 $\alpha > 2$ 时，导函数连续

作业

习题2-2: 2 (3,6, 8); 3 (2); 5; 6 (8);

7(1,8,9); 8(6,9); 10 (2); 11 (7)